

Finding a number from its percentage, and a part of it

The reverse question — given the part and the percentage, find the whole.

LESSON 59–62

4 × 40 MIN

MATEMATIKA 6, §13

S7 10

LESSON CLOCK

25'

35'

50'

40'

10

The three questions	25 min
Finding a percentage	35 min
Finding the whole	50 min
Expressing as a percentage	40 min
Homework	10 min

LEARNING OBJECTIVES

- Find a percentage of a quantity fluently.
- Find the whole when a percentage of it is given.
- Express one quantity as a percentage of another.
- Decide which of the three questions a problem is asking.

TEXTBOOK

National	pp. 167–178
Cambridge	Stage 7, pp. 96–104
Workbook	Exercise 10.2

01

Explanation

The three questions

QUESTION	GIVEN	METHOD	EXAMPLE
find a part	the whole and the percentage	multiply	15% of 240 = 36
find the whole	a part and its percentage	divide	36 is 15% $\implies$ 240
find the percentage	the part and the whole	$\frac{\text{part}}{\text{whole}} \cdot 100$	36 of 240 = 15%

THE SAME THREE NUMBERS, THREE WAYS ROUND

Every percentage problem in the course is one of these. Naming which one is being asked, before calculating anything, is the skill this block is really teaching.

Finding a percentage

TASK	WORKING	ANSWER
30%	$0.3 \cdot 250$	75
8%	$0.08 \cdot 250$	20
45%	$0.45 \cdot 250$	112.5
120%	$1.2 \cdot 250$	300

Every row is a percentage of 250.

TURN THE PERCENTAGE INTO A DECIMAL AND MULTIPLY

One multiplication replaces the two-step “find 1% then multiply”. Both are correct; the decimal method is faster once the conversion is automatic.

Finding the whole

If 36 is 15% of a number, then 1% is $36 \div 15 = 2.4$, so the whole is 240.

$$\text{whole} = \frac{\text{part}}{\text{percentage}} \times 100 \text{ or } \text{part} \div \text{the decimal}$$

GIVEN	1%	WHOLE
36	2.4	240
45	1.5	150
60	3	300
18	0.6	60

THE WHOLE IS LARGER THAN THE PART – UNLESS THE PERCENTAGE IS OVER 100

If 36 is 15% of a number, the number is 240, not 5.4. Multiplying when you should divide is the error, and comparing the two numbers exposes it.

Expressing as a percentage

PART	WHOLE	WORKING	PERCENTAGE
36	240	$\frac{36}{240} \cdot 100$	15%
42	56	$\frac{42}{56} \cdot 100$	75%
7	20	$\frac{7}{20} \cdot 100$	35%
45	36	$\frac{45}{36} \cdot 100$	125%

OVER 100% IS NOT A MISTAKE

The last row compares a larger quantity with a smaller one, so the answer exceeds 100%. It is wrong only when the part really is part of the whole.

02 Worked examples

EXAMPLE 1 Find 45% of 250.

1 $45\% = 0.45$

2 $0.45 \cdot 250$

3 $= 112.5$

ANSWER

112.5

EXAMPLE 2 36 is 15% of a number. Find the number.

- ① 1% is $36 \div 15 = 2.4$.
- ② 100% is 240.
- ③ Check: 15% of 240 is 36 ✓

ANSWER

240

EXAMPLE 3 What percentage of 56 is 42?

- ① $\frac{42}{56}$
The whole underneath.
- ② = 0.75
- ③ $\cdot 100 = 75\%$

ANSWER

75%

03 Interactive model

Give the three questions with the same three numbers — 36, 240 and 15% — and ask which is which; the pattern lands in one minute.

Which of the three questions?

“Find 15% of 240” needs:

multiplying

dividing

$\frac{\text{part}}{\text{whole}}$

nothing

“36 is 15% of what?” needs:

multiplying

dividing

adding

squaring

“36 out of 240 as a percentage” needs:

multiplying by 100

dividing then multiplying by 100

dividing by 100

nothing

30% of 250 is:

75

80

7.5

30

8% of 250 is:

2

20

31.25

8

45 is 30% of:

13.5

135

150

75

42 as a percentage of 56:

75%

133%

42%

14%

45 as a percentage of 36:

80%

125%

9%

45%

04

Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Percentage	Foiz	Процент
Whole	Butun	Целое
Part	Qism	Часть
To find the whole	Butunni topish	Найти целое
One per cent	Bir foiz	Один процент
Reverse question	Teskari masala	Обратная задача
To express as	Ifodalamoq	Выразить в
Check	Tekshirish	Проверка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

To find a percentage of a quantity you:

To find the whole from a part you:

36 is 15% of:

120% of 250 is:

270

300

320

250

7 as a percentage of 20:

7%

35%

14%

70%

A percentage over 100 means:

an error

the part exceeds the whole compared with

a fraction

a decimal

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *30% of 250*

ANSWER *75*

2. *8% of 250*

ANSWER *20*

3. *45% of 250*

ANSWER *112.5*

4. *120% of 250*

ANSWER *300*

5. *36 is 15% of*

ANSWER *240*

6. *45 is 30% of*

ANSWER *150*

7. *60 is 20% of*

ANSWER *300*

Medium · the standard question

7 PROBLEMS

1. 18 is 30% of

ANSWER 60

2. 36 as a percentage of 240

ANSWER 15%

3. 42 as a percentage of 56

ANSWER 75%

4. 7 as a percentage of 20

ANSWER 35%

5. 45 as a percentage of 36

ANSWER 125%

6. 12% of 4 500

ANSWER 540

7. 84 is 70% of

ANSWER 120

Hard · stretch and reason

7 PROBLEMS

1. 2.5% of 4 800

ANSWER 120

2. 150 is 12.5% of

ANSWER 1 200

3. A price of 54 000 is 90% of the original: the original

ANSWER 60 000

4. 64 as a percentage of 80

ANSWER 80%

5. If 30% of a class of 40 wear glasses, how many do not?

ANSWER 28

6. A tax of 12% adds 36 000: the price before tax

ANSWER 300 000

7. Why must the whole go underneath?

ANSWER A percentage compares the part with the whole it belongs to

07 Homework

Homework — 5 tasks

Name which of the three questions you are answering before you calculate.

1. Find 35% and 6% of 420.
2. 54 is 18% of a number. Find the number.
3. Write 27 as a percentage of 45.
4. Write 63 as a percentage of 45, and explain the answer.
5. A sum of 84 000 is 70% of a price. Find the price.